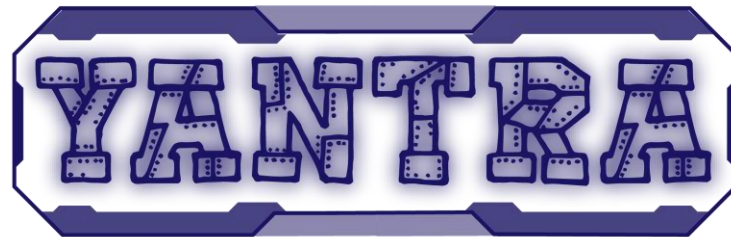


Laminated object manufacturing



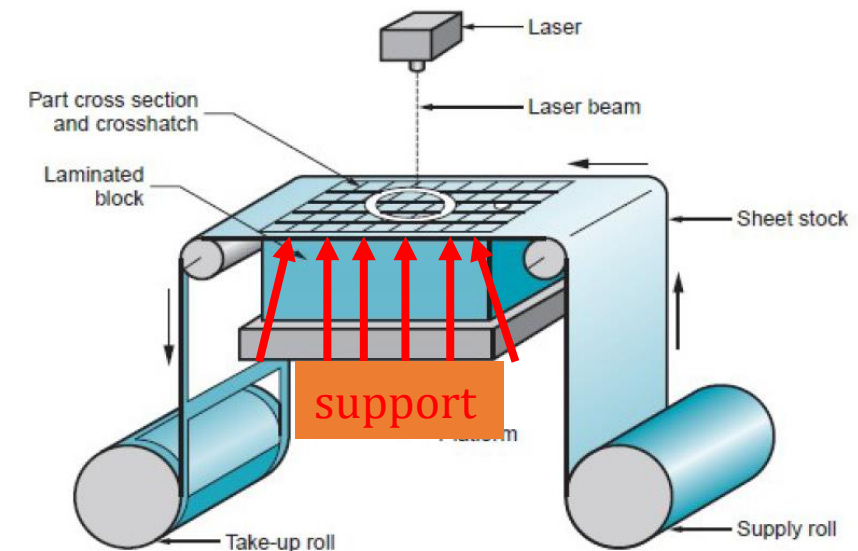
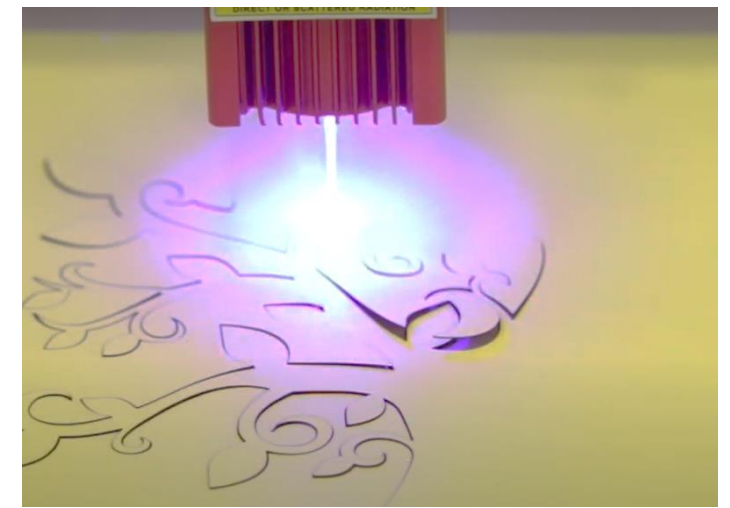
- Laminated Object Manufacturing (LOM), is a rapid prototyping (RP) technique that produces three-dimensional models with **paper, plastic, or composites**.
- LOM is actually more of a **hybrid between subtractive and additive processes**, in that the models are built up with layers of material, which are cut individually by a laser in the shape of the cross section of the part.
- Hence, as layers are being added, the excess material not required for that cross section is being cut away.
- LOM is one of the fastest RP processes for parts with larger cross-sectional areas, which makes it ideal for producing larger parts.

Principle

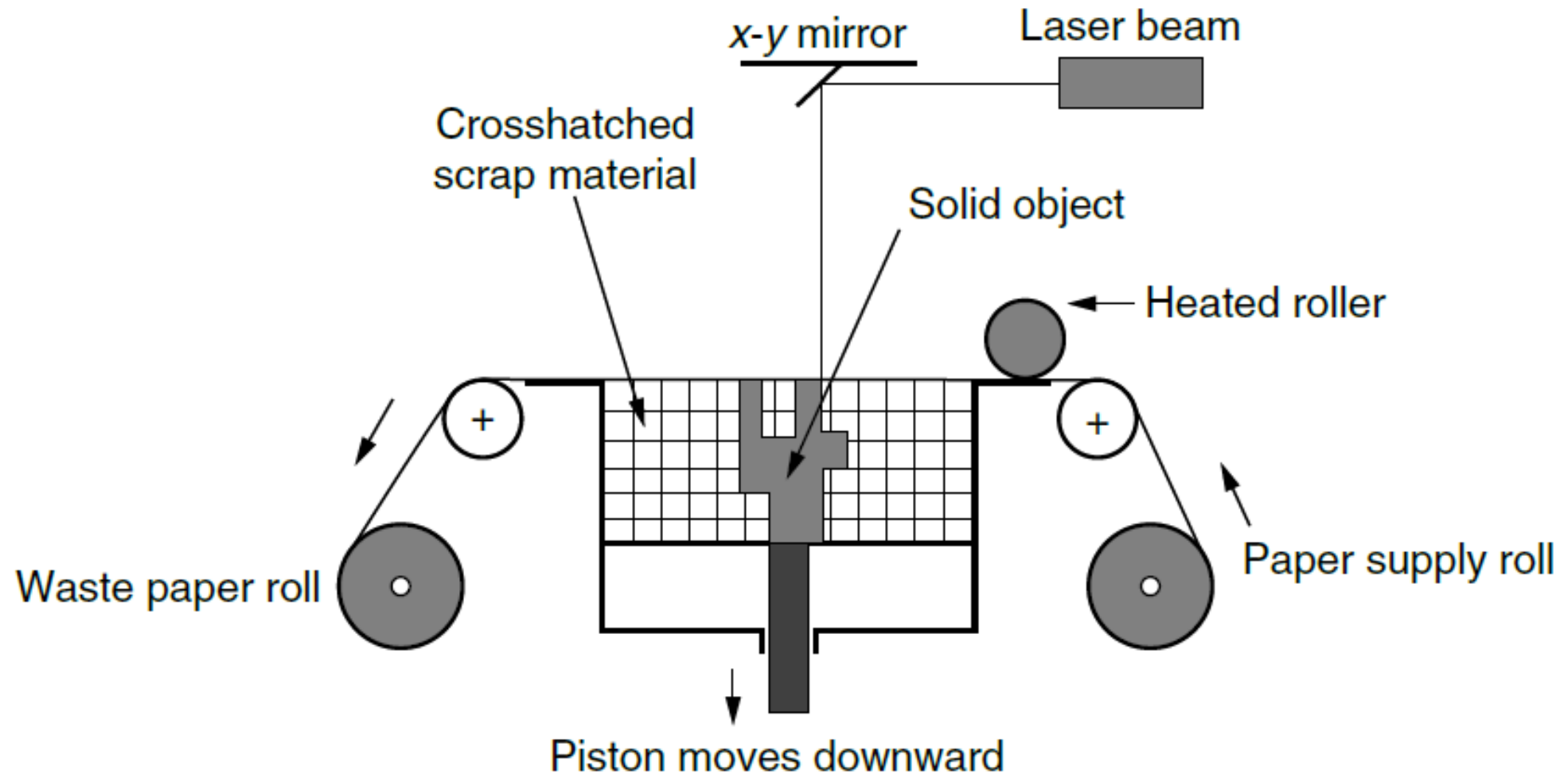
The LOM process is based on the following principles:

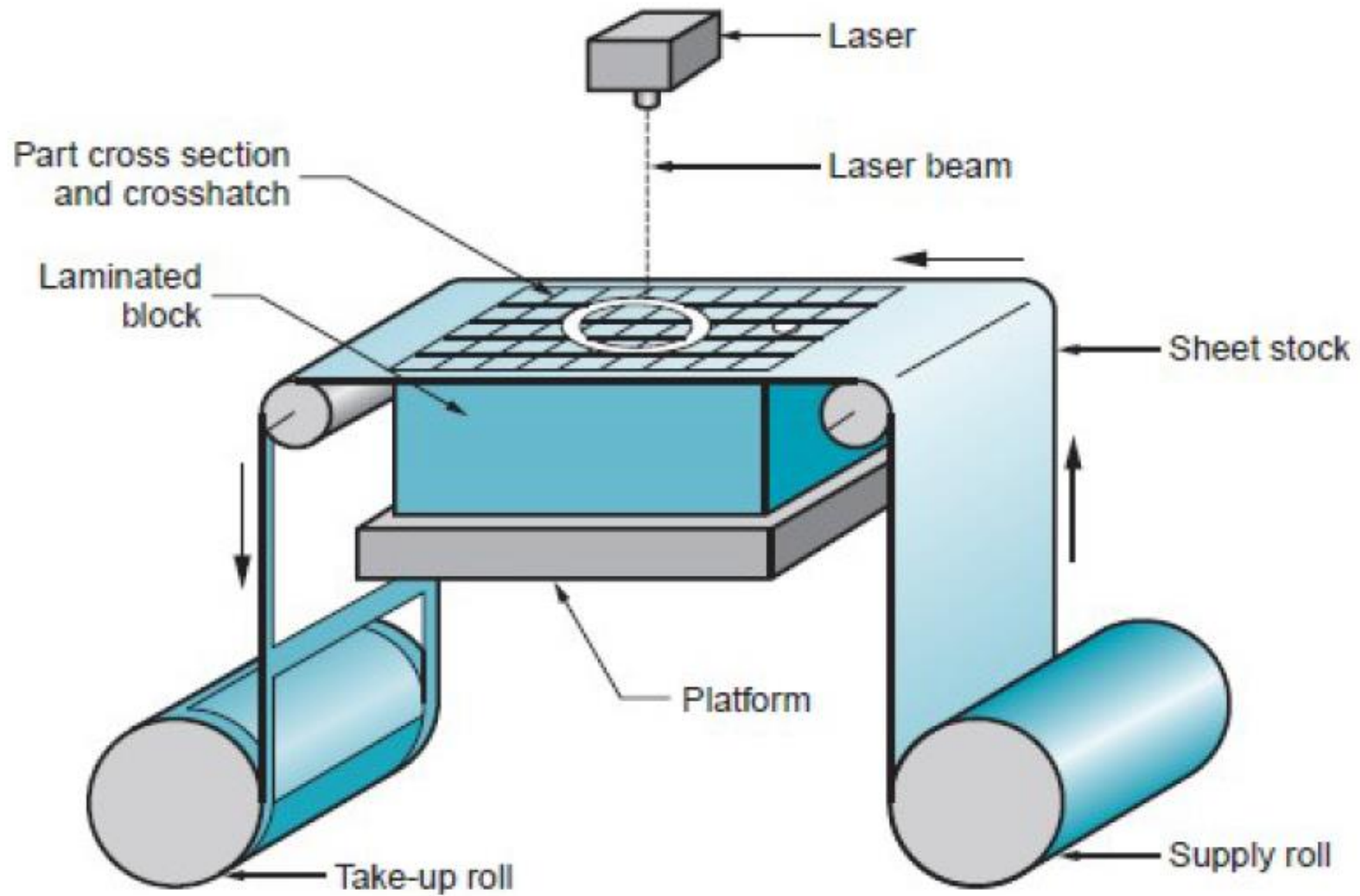
Parts are built, layer-by-layer, by laminating each layer of paper or other sheet-form materials and the contour of the part on that layer is cut by a CO₂ laser.

Each layer of the building process contains the cross-sections of one or many parts. The next layer is then laminated and built directly on top of the laser-cut layer.



Laminated object manufacturing





Process

LOM **doesn't require a preslice** of the STL file, that is, once the parameters are loaded into LOMSlice the STL file slices as the part builds.

This process of continuous slicing is called slice-on-the-fly.

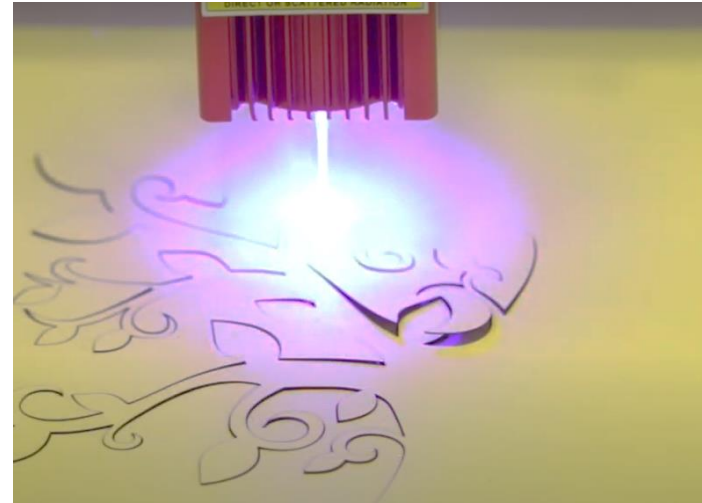
profiles of object cross sections are cut from paper or other web material using a laser.

The paper is **unwound** from a feed roll onto the stack

and first bonded to the previous layer using a heated roller

which **melts** a plastic coating on the bottom side of the paper.

The profiles are then traced by an **optics system** that is mounted to an x-y stage.



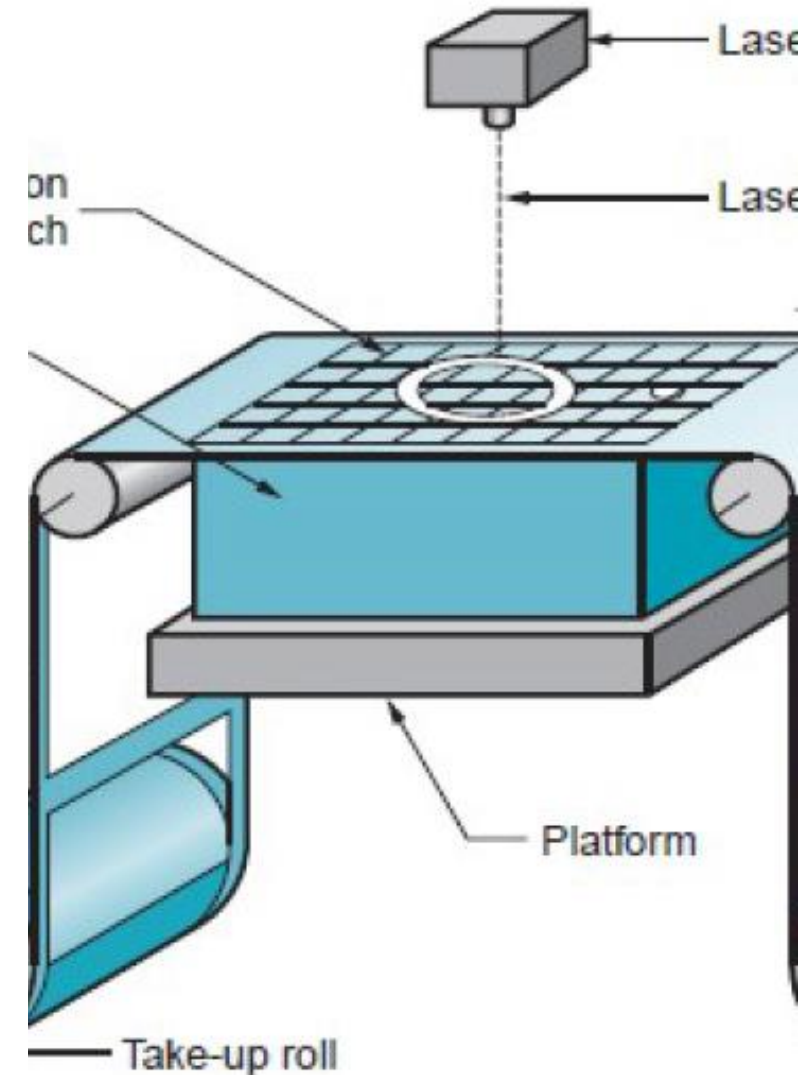
After the laser has cut each profile, the roll of paper is advanced, a new layer is **glued** onto the stack, and the process is repeated.

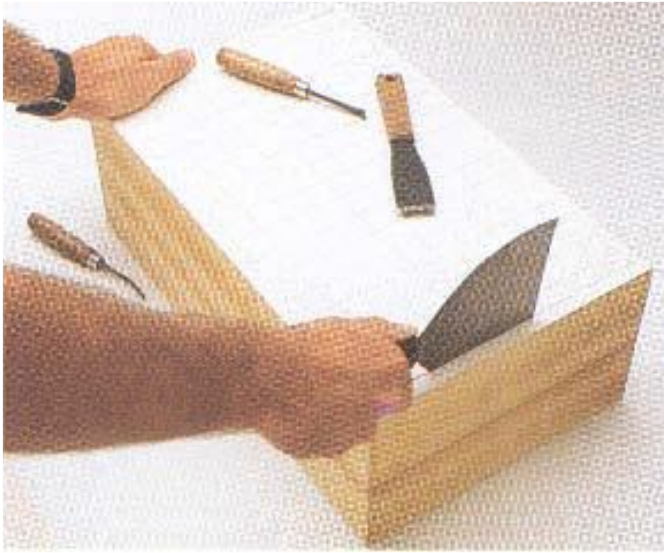
After fabrication, some **trimming, hand finishing, and curing** are needed.

The method is **self-supporting** for overhangs and undercuts.

Areas of cross sections, which are to be removed in the final object are **heavily crosshatched** with the laser **to facilitate removal**

and convert these excess materials into a support structure.





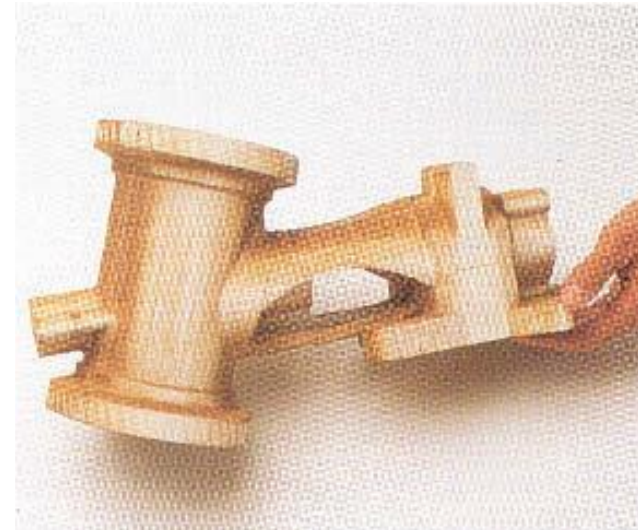
1. The laminated stack is removed from the machine's elevator plate.



2. The surrounding wall is lifted off the object to expose cubes of excess material.



3. Cubes are easily separated from the object's surface.



4. The object's surface can then be sanded, polished or painted, as desired.

In LOM the surface finish, accuracy, and stability of paper objects are **not as good as** for materials used with other RP methods.

However, material **cost** is very **low**, and objects have the look and feel of wood and can be worked and finished in the same manner to form patterns for sand castings.

While there are **limitations on materials**, work has been done with plastics, composites, ceramics, and metals.

Some of these materials are available on a limited commercial basis.

Advantages

1. **Wide variety of materials:** These include a wide variety of organic and inorganic materials such as **paper, plastics, metals, composites and ceramics**. Commercial availability of these materials allow users to vary the type and thickness of manufacturing materials to meet their functional requirements and specific applications of the prototype.
2. **Fast build time:** The laser in the LOM process **does not scan the entire surface** area of each cross-section, rather it **only outlines its periphery**. Therefore, parts with thick sections are produced just as quickly as those with thin sections, making the LOM process especially advantageous for the production of large and bulky parts.

4. Support structure:

There is **no need** for additional support structure as the part is supported by its own material that is outside the periphery of the part built.

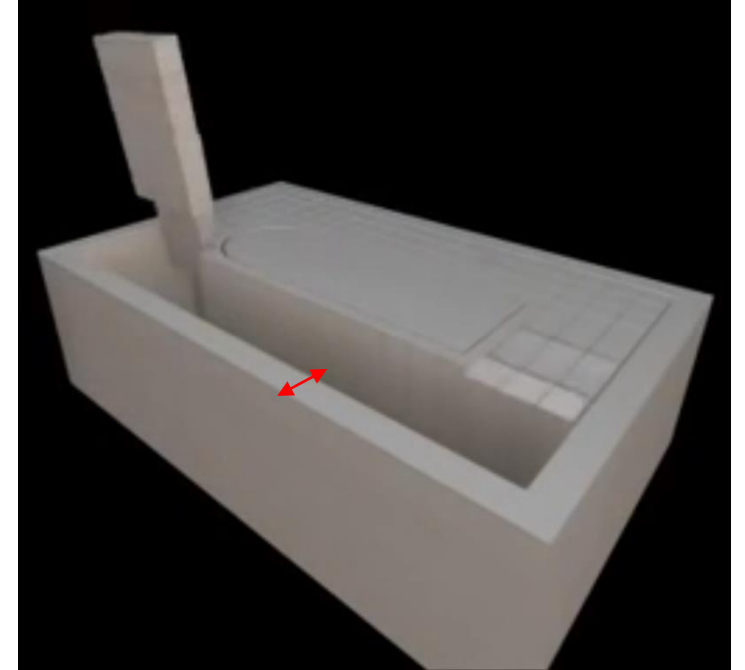
These are not removed during the LOM process and therefore automatically act as supports for its delicate or overhang features.

5. Post-curing:

The LOM process **does not need** to convert expensive, and in some cases toxic, liquid polymers to solid plastics or plastic powders into sintered objects. Because sheet materials are not subjected to either physical (or) chemical phase changes, the finished LOM parts do not experience warpage, internal residual stress, or other deformations.

Disadvantages

1. **Precise power adjustment:** The power of the laser used for cutting the perimeter (and the crosshatches) of the prototype needs to be precisely controlled so that **the laser cuts only the current layer of lamination and not penetrate into the previously cut layers**. Poor control of the cutting laser beam may cause distortion to the entire prototype.
2. **Fabrication of thin walls:** The LOM process is not well suited for building parts with delicate thin walls, especially in the Z-direction. This is because such walls usually are not sufficiently rigid to withstand the post-processing process when the cross-hatched outer perimeter portion of the block is being removed.



- 3. **Integrity of prototypes:**
- The part built by the LOM process is essentially held together by the heat sealed adhesives. The **integrity** of the part is therefore entirely dependent on the **adhesive strength** of the glue used, and as such is limited to this strength. Therefore, parts built may not be able to withstand the vigorous mechanical loading that the functional prototypes may require.
- 4. **Removal of supports:**
- The most labor-intensive part of the process separating its support material within the rectangular block of laminated material.
- This is usually done with wood carving tools and can be tedious and time consuming.
- The person working during this phase needs to be careful and aware of the presence of any delicate parts within the model so as not to damage it.



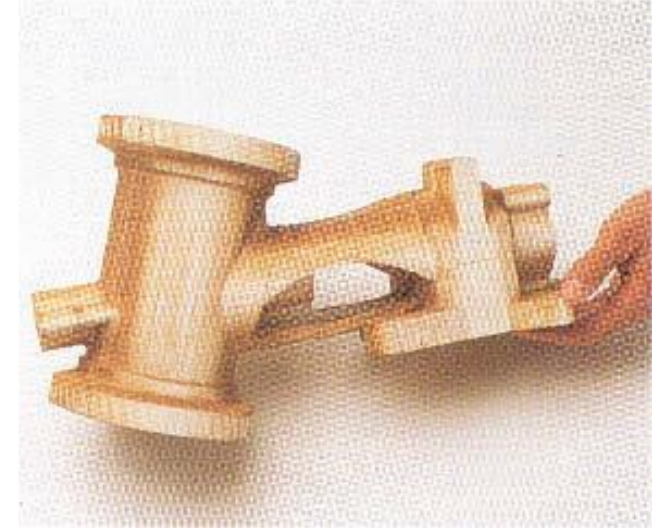
Applications

- **Visualization:**

Many companies utilize LOM's ability to produce exact dimensions of a potential product purely for visualization.

LOM part's **wood-like** composition allows it to be **painted** or finished as a true replica of the product.

As the LOM procedure is inexpensive several models can be created, giving sales and marketing executives opportunities to utilize these prototypes for consumer testing, marketing product introductions, packaging samples, and samples for vendor quotations.



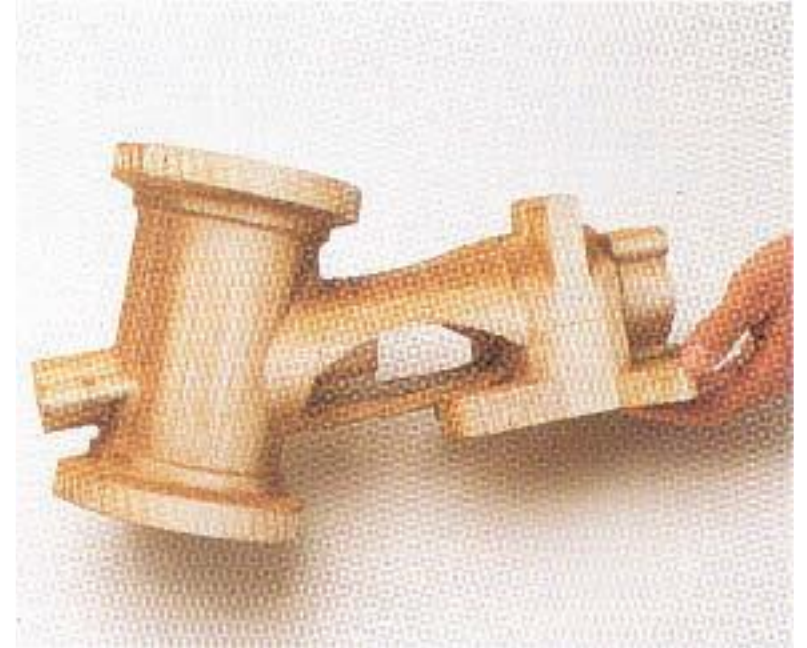
**wood-like
composition**

Form, fit and function:

- LOM parts lend themselves well for **design verification** and **performance evaluation**.
- In low-stress environments LOM parts can withstand basic tests, giving manufacturers the opportunity to make changes as well as evaluate the aesthetic property of the prototype in its total environment.

- **Manufacturing:**

The LOM part's composition is such that, based on the sealant or finishing products used, it can be further tooled for use as a pattern or mold for most secondary tooling techniques including: investment casting, casting, sanding casting, injection molding, silicon rubber mold, vacuum forming and spray metal molding.



pattern

LOM parts offer several advantages important for the secondary tooling process, namely: predictable level of accuracy across the entire part; stability and resistance to shrinkage, warpage and deformity; and the flexibility to create a master or a mold.

In many industries the master created through secondary tooling, or even when the LOM part serves as the master (e.g., vacuum forming), withstands enough injections, wax shootings or vacuum pressure to produce a low production run from 5 to 1000 pieces.